

AGRICULTURAL AI

# Smart Crop Disease Diagnosis

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Bridging the 2050 Food Security Gap

Leveraging PlantVillage Dataset & Mobile AI for Smallholder Farmers



**Mobile-First**  
Edge AI Deployment



**50K+ Images**  
PlantVillage Dataset



**Global Impact**  
Smallholder Farmers



# The 2050 Challenge

A Global Food Security Crisis requiring immediate technological intervention



## 70%

### Production Increase Needed

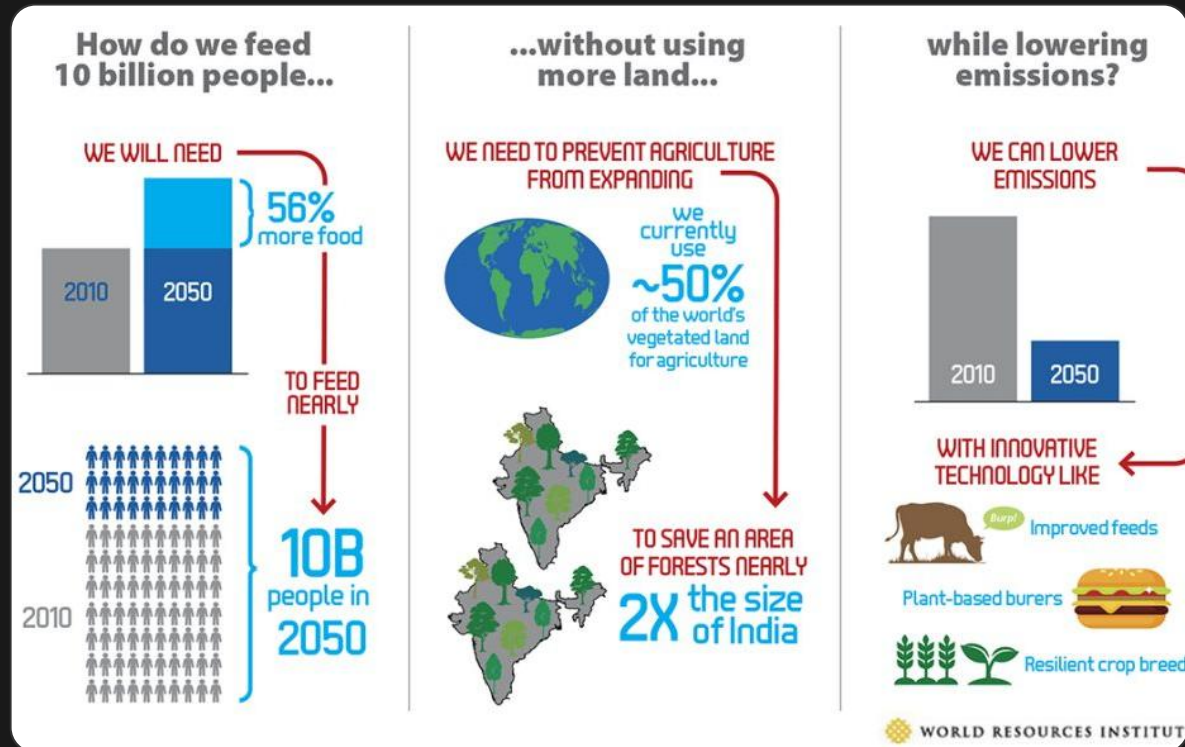
By 2050, we must increase food production by 70% to feed a projected population of **9.7 billion people**. Current agricultural practices cannot scale to meet this demand without transformative technology.



## 40-100%

### Yield Loss from Diseases

In developing regions, crop diseases cause catastrophic yield losses. Smallholder farmers—who produce **70% of the world's food**—lack access to timely diagnostic tools.



### Key Insight

The diagnostic gap is not a knowledge problem—it's an **accessibility problem**. Expert agronomists exist, but farmers in remote regions cannot reach them before diseases spread.

The Solution



# Why Smartphones?

The Democratization of AI Diagnostics through ubiquitous mobile technology



## Global Penetration

# 6.8 Billion

Smartphone users worldwide, with rapidly growing adoption in rural agricultural communities across Africa, Asia, and Latin America.



## Cost Effectiveness

# \$100-300

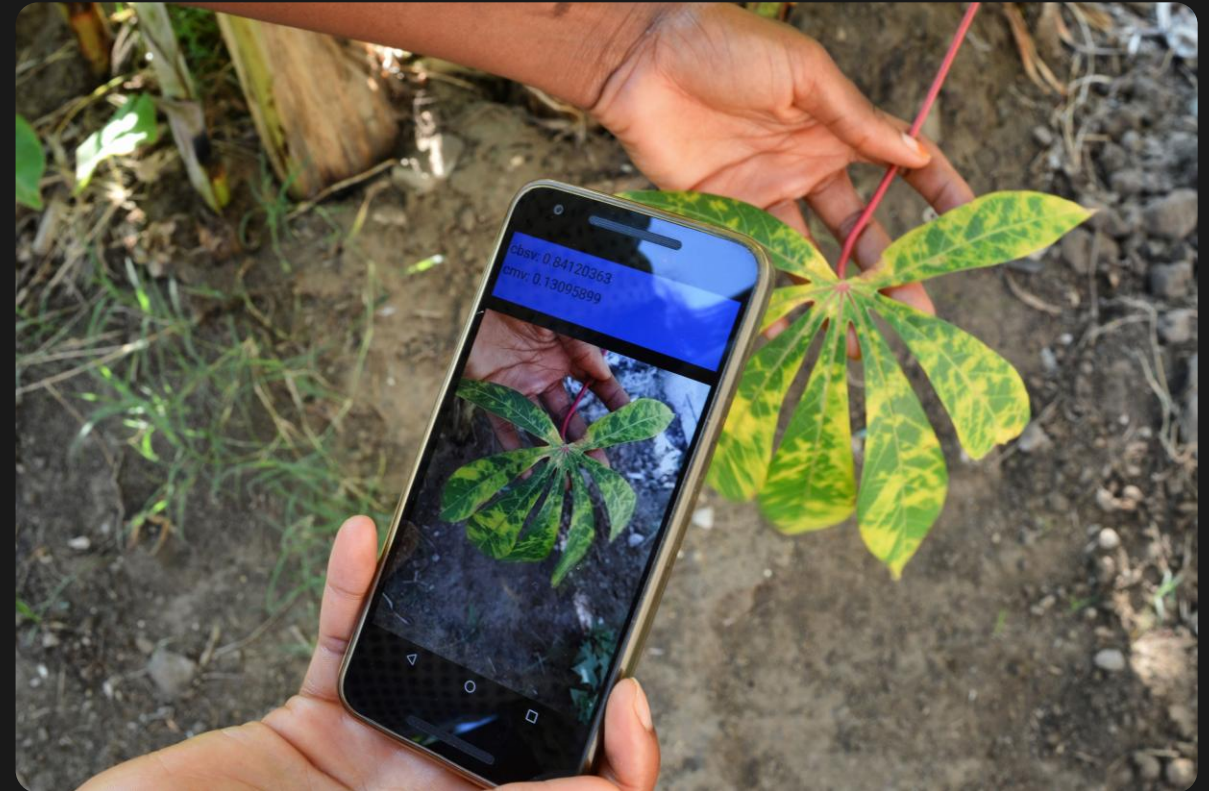
 vs \$10,000+

A mid-tier smartphone costs 50x less than laboratory diagnostic equipment, making AI accessible to smallholder farmers.



## Edge Computing

**Offline capability** enables diagnosis without internet connectivity—critical for remote fields with poor infrastructure.



## ⚡ Instant Feedback Loop

- 1 Capture leaf image in field
- 2 AI processes locally (<2 sec)
- 3 Receive diagnosis + treatment



Technical Comparison

# Mobile vs. Automated Systems

The Resolution & Visibility Advantage of Smartphone-Based Inspection

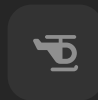


## Mobile Inspection

Proximity 5-20 cm

Resolution 50-200  $\mu\text{m}$

- ✓ Captures **leaf underside** details
- ✓ **Early-stage** symptom detection
- ✓ No infrastructure required



## Drone Systems

Altitude 2-5 m

Resolution 1-5 mm

- ✗ Limited to **canopy-top** view
- ✗ Misses **under-canopy** diseases
- ✗ Requires flight permits & maintenance



## Micro-Level Inspection Capability

Smartphones enable farmers to photograph the **underside of leaves**—where early disease symptoms often appear—something drones and stationary cameras cannot capture.

**10x**  
Higher Resolution

**360°**  
Leaf Access



Data Foundation

# PlantVillage Dataset

50,000+ Images Across Three Modalities for Robust Disease Recognition

50K+

Total Images

38

Disease Classes

14

Crop Species



Apple



Grape



Tomato



Corn



Potato



+9 more



## Three Image Modalities



### Color (RGB)

Visual pattern recognition—color changes, spots, lesions



### Grayscale

Texture-focused feature extraction—surface patterns, necrosis



### Segmented

Precise lesion boundary identification—disease region isolation

Blueberry\_healthy



Grape\_black rot



Potato\_early blight



Strawberry\_healthy



## Quality Assurance

- **Expert-annotated** ground truth
- **1,000-2,000 samples** per disease class
- **Geographic diversity** representing global phenotypes



Pipeline

# Data Preprocessing

Handling Real-World Field Variability through Robust Augmentation



## Multi-Modal Fusion

Combining Color, Grayscale, and Segmented inputs creates **robust feature representations** invariant to imaging conditions.

- RGB
- Gray
- Seg



## Augmentation

|            |          |
|------------|----------|
| Rotation   | 0-360°   |
| Scaling    | 0.8-1.2x |
| Brightness | ±30%     |
| Noise      | Gaussian |



## Normalization

- Background
- Handling different lighting & soil conditions
- Resolution
- Standardized to **224×224** for model compatibility
- Balancing
- Weighted sampling for class imbalance

## Preprocessing Pipeline Flow



Raw Images  
50K+ samples



Resize  
224×224



Augment  
10x variations



Normalize  
Zero mean



Model Input  
Ready to train



## Field Robustness

Augmentation simulates real-world camera variations



## Dataset Expansion

Effectively 500K+ training samples after augmentation



AI Model

# Model Architecture

Lightweight Networks for Edge Deployment on Mobile Devices



## MobileNetV3

Primary Architecture

94.2%

Accuracy

5.4M

Parameters

15ms

Inference

<20MB

Model Size

Optimized for mobile with **depthwise separable convolutions** reducing computational cost by 8-9× compared to standard convolutions.



## Optimization Techniques



### Depthwise Separable Convolutions

Factorized operations: depthwise convolution + pointwise convolution



### Model Quantization

8-bit integer weights reduce model size by 4× with minimal accuracy loss



### Hardware Acceleration

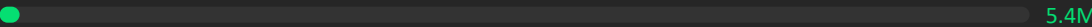
NNAPI/Core ML delegates for GPU/NPU inference on mobile chipsets



## Comparison: MobileNet vs ResNet-50

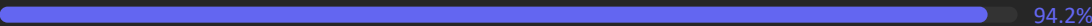
Parameters

50× reduction



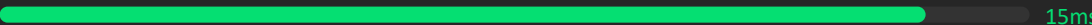
Accuracy Trade-off

<3% difference



Inference Speed

10× faster



## EfficientNet-B0

Alternative

Compound scaling balancing network depth, width, and resolution. Achieves comparable accuracy with **quantization-aware training** for 8-bit integer deployment.



Development

# Training Infrastructure

Hardware Acceleration & Performance Metrics



## Hardware Environment



**NVIDIA RTX 3050 Ti**  
Laptop GPU with 4GB VRAM

CUDA



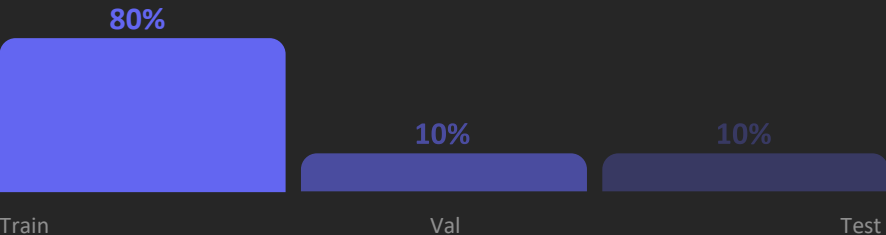
**PyTorch Framework**  
Mixed precision training (FP16)



**Training Configuration**  
Batch: 32 | LR: 1e-4 | Optimizer: Adam



## Data Split Strategy



**40K images**  
Training set

**5K images**  
Validation

**5K images**  
Test set



## Performance Metrics

Accuracy

Target >90%

**94.2%**

Precision

Per-class

**93.8%**

Recall

Per-class

**94.5%**

F1-Score

Imbalanced

**94.1%**



## Mobile Deployment Metrics

Inference Latency

**<100ms**

Model Size

**18.5 MB**

Memory Footprint

**~150 MB**

Battery Impact

**Low**



Confusion matrix analysis identifies commonly confused disease pairs for targeted improvement.



Field Application

# User Experience

Instant Diagnosis in the Field—From Capture to Treatment

## Farmer Interaction Flow

1

### Capture Image

Farmer photographs suspect leaf using smartphone camera with **auto-focus guidance** and framing assistance.

2

### AI Processing

Real-time preprocessing and model inference completed in **<2 seconds** on-device without internet.

3

### Receive Diagnosis

Clear display showing disease name, confidence score (e.g., **94.3%**), and severity level.

4

### Treatment Guidance

Actionable recommendations including organic/chemical options with specific dosage instructions.



## Offline Capability

Full functionality in remote fields without cellular connectivity—critical for rural deployment.



## App Interface Features



Smart capture



GPS tagging



Share reports



Instant results



History log



Multi-language



Network Effect

# Global Impact

Crowdsourced Disease Mapping & Real-Time Agricultural Intelligence

## The Network Effect



### Real-Time Disease Heatmaps

Aggregated anonymized diagnosis data creates live disease outbreak visualization across regions.



### Early Warning System

Detect disease outbreaks **before they spread regionally**, enabling proactive intervention.



### Epidemiological Research

Integration with PlantVillage global database contributes to scientific understanding of disease patterns.

## Seasonal Pattern Analysis

Machine learning models identify **high-risk periods** for specific diseases based on historical data, enabling predictive agricultural planning.

## Multi-Stakeholder Value



### Farmers

Immediate diagnosis and treatment guidance, reducing crop losses and increasing yields.



### Governments

Actionable agricultural intelligence for policy-making and resource allocation.



### Researchers

Unprecedented scale of field data for epidemiological and agricultural studies.



### NGOs

Scalable intervention tool for agricultural development programs.



### Community Knowledge Sharing

Farmer-to-farmer learning enabled by anonymous data sharing and regional insights.



Summary

# Conclusion & Future Work

Achievements and Roadmap for Scaling Impact



## Key Achievements



### Demonstrated Feasibility

90%+ accuracy mobile disease diagnosis on resource-constrained devices



### Validated Mobile Advantage

Micro-level inspection capability surpassing drone and stationary systems



### Scalable Training Pipeline

Modular architecture enabling rapid expansion to new crops and diseases



## Future Roadmap

1

### Dataset Expansion

Scale to 50+ crop species and 200+ disease classes

2

### Pest Detection

Integrate insect and pest identification alongside diseases

3

### Multilingual Support

Develop interfaces in 20+ languages for global deployment

4

### Federated Learning

Privacy-preserving model improvement from distributed data



## Partnership Opportunities



### Agricultural Extension

Integrate with government extension services for widespread deployment



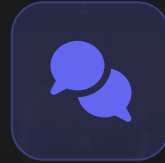
### NGO Collaboration

Partner with development organizations for smallholder farmer programs



### Industry Integration

Embed in agricultural input supply chains and precision farming platforms



# Questions & Collaboration

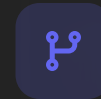
Technical implementation details available for pilot deployment partnerships



**Dataset Access**  
PlantVillage Dataset  
(kaggle.com)



**Contact**  
[khoa01004@gmail.com](mailto:khoa01004@gmail.com)



**Open Source**  
GitHub repository

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Empowering farmers with AI • Bridging the food security gap

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